Introduction to Algorithms

Algorithms:

- Step by step method of solving problem.
- The word – “Algorithm” derives from the name of the 9th Century Persian Mathematician – “Al – khowarizimi”.
- Step by step means one step is followed by the other step. i.e. Previous step is pre – requisite of the next step.

Properties – An Algorithm must have:

- Input
- Output
- Precision
- Determinism
- Finiteness
- Correctness
- Generality

Input & Output

- An algorithm must have something as Input.
- It should also give some as Output

Precision

- Steps, written in the algorithm, should be precisely stated.

Determinism

- The intermediate results of each step, of execution, are unique and are determined only by inputs and results of the preceding steps.

Finiteness

- Algorithm should terminate.
- It means the algorithm should stop after finitely many instructions have been executed.

Correctness

- Output produced by the Algorithm should be correct.

Generality

- Algorithm should be generalized in nature.
- It should apply to a set of Input.

Example

- To find the Largest among three Numbers (a,b,c)

    1. x = a
    2. If b > x then x = b
    3. If c > x then x = c
    4. Print x

  The above procedure (4 steps) is perfect example algorithm. It has all properties.

Big Oh

- Let f and g be non negative function on the +ve integers.
- We write $f(n) = O(g(n))$

  i.e. f(n) is of order, at most g(n)

  or

  f(n) is *big oh* of g(n)

  If there exist constants C_1 and N_1 such that $f(n) \leq C_1\, g(n)$ for all $n \geq N_1$

  i.e. f is bounded above by g

  or g is an asymptotic upper bound for f

Omega (Ω)

- Let f and g be non negative function on the +ve integers.
- We write $f(n) = \Omega(g(n))$

  i.e. $f(n)$ is of order, at least $g(n)$

  or

  $f(n)$ is *omega* of $g(n)$

  If there exist constants C_2 and N_2 such that $f(n) \geq C_2\, g(n)$ for all $n \geq N_2$

  i.e. f is bounded below by g

  or g is an asymptotic lower bound for f